**SAPTHAGIRI COLLEGE OF ENGINEERING, BENGALURU-560 057.**
**(Affiliated to VTU, Belagavi & Approved by AICTE, New Delhi)**
**DEPARTMENT OF MATHEMATICS**
**QUESTION BANK (I SEM)**
**ACADEMIC YEAR: 2023-24 (ODD SEMESTER)**

## MODULE-1: DIFFERENTIAL CALCULUS-1

**SUB: MATHEMATICS-1** **SUB CODE: BMATE/C/S/M101**

1. With usual notation prove that $tan\emptyset = r\left(\frac{d\theta}{dr}\right)$

2. With usual notation prove that $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4}\left[\frac{dr}{d\theta}\right]^2$

3. Find the angle between radius vector and tangent of the curve
   i. $r^m = a^m(\cos m\theta + \sin m\theta)$ at $\theta = \pi/2$
   ii. $r = a(1+\sin\theta)$ $\theta = \pi/2$

4. Find the angle between the polar cures
   i. $r = a\log\theta$ $r = \frac{a}{\log\theta}$
   ii. $r^2\sin 2\theta = 4$ $r^2 = 16\sin 2\theta$
   iii. $r = \frac{a}{1+\cos\theta}$ $r = \frac{b}{1-\cos\theta}$
   iv. $r = a(1+\cos\theta)$ $r = b(1-\cos\theta)$

5. Show that the following pairs of curves intersect each other orthogonally.
   i. $r = a(1+\sin\theta)$ and $r = b(1-\sin\theta)$
   ii. $r^n = a^n\cos n\theta$ and $r^n = b^n\sin n\theta$
   iii. $r = a(1+\cos\theta)$ and $r = b(1-\cos\theta)$
   iv. $r(1+\cos\theta) = a$ and $r(1-\cos\theta) = b$

6. Find the pedal equation of the curve
   i. $r^n = a^n\cos n\theta$

ii. $2a/r = (1+\cos\theta)$

iii. $r^m = a^m(\cos m\theta + \sin m\theta)$

iv. $r = ae^{\theta\cot\alpha}$

7. Establish the pedal equation of the curve $r^n = a^n \sin n\theta + b^n \cos n\theta$ in the form $p^2(a^{2n} + b^{2n}) = r^{2n+2}$
8. Find the radius of curvature of the following curves

i. $a^2 y = x^3 - a^3$ at the point where the curve cuts the x-axis

ii. $x^3 + y^3 = 3axy$ at the point (3a, 3a)

iii. $y^2 = \dfrac{a^2(a-x)}{x}$ at the point (a, 0)

iv. $x^4 + y^4 = 2$ at the point (1, 1)

v. $x = a(t + \sin t)$, $y = a(1-\cos t)$

vi. $x = a\log(\sec t + \tan t)$, $y = a\sec t$

vii. $r = a(1+\cos\theta)$

viii. $\dfrac{2a}{r} = 1 + \cos\theta$

ix. $r(1-\cos\theta) = 2a$

9. For the curve $y = \dfrac{ax}{a+x}$ show that $\left(\dfrac{2\rho}{a}\right)^{2/3} = \left(\dfrac{x}{y}\right)^2 + \left(\dfrac{y}{x}\right)^2$

10. If $\rho$ be the radius of curvature at any point P(x, y) on the parabola $y^2 = 4ax$, Show that $\rho^2$ varies as $(SP)^3$ where S is the focus of the parabola.

11. Derive an expression for Radius of curvature in case of a Cartesian curve.

12. Derive an expression for Radius of curvature in case of a Polar Curve.